

AMAZON BRAZIL SALES DATA ANALYSIS

Strategic Insights from Amazon Brazil for Amazon India

SQL Milestone Project

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Course NextLeap Data Analytics Fellowship

Tools Used PostgreSQL • Microsoft Excel

1 Average Payment Value by Payment Type

BUSINESS CONTEXT

Amazon India needs standardized average payment values by method to support consistent, comparable financial reporting.

KEY FINDINGS

- Credit card holds the highest average transaction value at 163 BRL.
- Boleto (145 BRL) and debit card (143 BRL) trail closely behind.
- Voucher averages just 66 BRL, the lowest of all methods.

BUSINESS INSIGHT

- Credit cards consistently support higher-value transactions, indicating stronger customer willingness to finance larger purchases.
- Voucher's low average suggests it functions as a supplementary discount tool rather than a primary payment method.

RECOMMENDATION

- Design credit- and EMI-linked offers to lift average order value on card transactions.
- Evaluate whether India's payment mix shows a similar credit-driven ticket-size pattern before scaling the offer.

```
Query  Query History
1  -- Calculate the average payment value for each payment type.
2  -- Round the average payment value to the nearest integer
3  -- and display the results in ascending order.
4
5  SELECT
6      payment_type,
7      ROUND(AVG(payment_value), 0) AS rounded_avg_payment
8  FROM amazon_brazil.payments
9  GROUP BY payment_type
10 ORDER BY rounded_avg_payment;
```

Data Output Messages Notifications

	payment_type character varying (30)	rounded_avg_payment numeric
1	not_defined	0
2	voucher	66
3	debit_card	143
4	boleto	145
5	credit_card	163

Order Distribution by Payment Type

BUSINESS CONTEXT

Understanding how orders split across payment methods guides checkout investment and payment-partner prioritization.

KEY FINDINGS

- Credit card accounts for 73.9% of all orders, the clear dominant method.
- Boleto follows distantly at 19.0%, with voucher at 5.6%.
- Debit card (1.5%) and not-defined (0.0%) are negligible.

BUSINESS INSIGHT

- Near-total reliance on one method makes checkout uptime and card-processing reliability business-critical for revenue continuity.
- Brazil's card-first mix is unlikely to translate directly to India, where alternative digital rails are typically dominant.

RECOMMENDATION

- Evaluate India's actual payment-method distribution before assuming a card-first checkout design will convert best.
- Build redundancy across at least two leading payment rails to avoid the single-point-of-failure risk seen here

```
Query  Query History
14
15 SELECT
16     payment_type,
17     ROUND(
18         COUNT(*) * 100.0 / (SELECT COUNT(*) FROM amazon_brazil.payments),
19         1
20     ) AS percentage_orders
21 FROM amazon_brazil.payments
22 GROUP BY payment_type
23 ORDER BY percentage_orders DESC;
24
25
26
27
28
29
```

Data Output Messages Notifications

	payment_type character varying (30)	percentage_orders numeric
1	credit_card	73.9
2	boleto	19.0
3	voucher	5.6
4	debit_card	1.5
5	not_defined	0.0

'Smart' Products Priced 100–500 BRL

BUSINESS CONTEXT

Amazon India needs a target list of mid-priced 'Smart' products to support promotional bundling and paid placement.

KEY FINDINGS

- The highest-priced match reaches 439.99 BRL.
- Repeat SKUs cluster densely near 299.90–329.90 BRL.
- Several identical product IDs recur at the same fixed price point.

BUSINESS INSIGHT

- The 300–350 BRL cluster signals a market 'sweet spot' where smart-product demand concentrates.
- Stable, non-discounted repeat pricing indicates these SKUs sell reliably without promotional support.

RECOMMENDATION

- Anchor a 'Smart Devices' shelf in India around the equivalent of this mid-price band.
- Confirm the price sensitivity holds locally before scaling festive placement around these SKUs.

```
Query  Query History
24
25 -- Find products priced between 100 and 500 BRL
26 -- whose category name contains the word 'smart'.
27 -- Display the product ID and price in descending order.
28
29 SELECT
30     oi.product_id,
31     oi.price
32 FROM amazon_brazil.order_items oi
33 INNER JOIN amazon_brazil.product p
34 ON oi.product_id = p.product_id
35 WHERE oi.price BETWEEN 100 AND 500
36 AND LOWER(p.product_category_name) LIKE '%smart%'
37 ORDER BY oi.price DESC;
38
39
```

Data Output Messages Notifications

	product_id character varying (50)	price numeric (10,2)
1	1df1a2df8ad2b9d3aa49fd851e314...	439.99
2	7debe59b10825e89c1cbcc8b190c...	349.99
3	ca86b9fe16e12de698c955aedff0a...	349.00
4	ca86b9fe16e12de698c955aedff0a...	349.00
5	0e52955ca8143bd179b311cc454a...	335.00
6	7aeaa8f3e592e380c420e8910a717...	329.90
7	7aeaa8f3e592e380c420e8910a717...	329.90
8	7aeaa8f3e592e380c420e8910a717...	329.90
9	7aeaa8f3e592e380c420e8910a717...	329.90
10	7aeaa8f3e592e380c420e8910a717...	329.90
11	7aeaa8f3e592e380c420e8910a717...	329.90
12	d1b571cd58267d8cac8b2afd6e288...	299.90
13	d1b571cd58267d8cac8b2afd6e288...	299.90
14	66ffe28d0fd53808d0535eee4b90a...	254.00
15	f06796447de379a26dde5fcac6a1a...	239.90
16	d3d5a1d52abe9a7d234908d873fc...	229.90

Top 3 Months by Total Sales Value

BUSINESS CONTEXT

Identifying peak-selling months enables Amazon India to plan inventory, staffing and marketing spend around demand spikes.

KEY FINDINGS

- May is the strongest month at 1,746,901 BRL in total sales.
- August follows at 1,696,822 BRL.
- July rounds out the top three at 1,658,924 BRL.

BUSINESS INSIGHT

- Peak demand spans a sustained May–August window rather than one holiday spike, pointing to a multi-month seasonal driver.
- This diverges from the typical Nov–Dec festive peak, suggesting category mix — not calendar events — drives the pattern.

RECOMMENDATION

- Map India's own mid-year demand drivers against this Brazil pattern before locking inventory plans.
- Pressure-test whether a comparable multi-month peak exists locally before committing marketing spend.

```
Query  Query History
38
39  -- Determine the top 3 months with the highest total sales value.
40  -- Round the total sales to the nearest integer.
41
42  SELECT
43      EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month,
44      ROUND(SUM(p.payment_value), 0) AS total_sales
45  FROM  amazon_brazil.orders o
46  INNER JOIN amazon_brazil.payments p
47  ON o.order_id = p.order_id
48  GROUP BY EXTRACT(MONTH FROM o.order_purchase_timestamp)
49  ORDER BY total_sales DESC
50  LIMIT 3;
51
52
53
```

Data Output Messages Notifications



	month numeric	total_sales numeric
1	5	1746901
2	8	1696822
3	7	1658924

Categories with Largest Price Spread (>500 BRL)

BUSINESS CONTEXT

Wide within-category price ranges can signal coexisting budget and premium demand, informing tiered assortment planning.

KEY FINDINGS

- utilidades_domesticas leads with a 6,731.94 BRL spread, followed by pcs and artes.
- Electronics-adjacent categories (eletroportateis, instrumentos_musicais) also show 4,000+ BRL spreads.
- 18 categories in total exceed the 500 BRL spread threshold.

BUSINESS INSIGHT

- Categories spanning budget-to-premium tiers are under-served by flat, one-size pricing.
- High-spread categories are strong candidates for a structured 'Good-Better-Best' merchandising approach.

RECOMMENDATION

- Pilot tiered product ladders (entry, mid, premium) in India's highest-spread category equivalents.
- Validate that similar dual-tier demand exists locally before extending the structure catalog-wide.

Query Query History

```
52
53 -- Find product categories where the difference between
54 -- the maximum and minimum product prices is greater than 500 BRL.
55
56 SELECT
57     p.product_category_name,
58     MAX(oi.price) - MIN(oi.price) AS price_difference
59 FROM amazon_brazil.product p
60 INNER JOIN amazon_brazil.order_items oi
61 ON p.product_id = oi.product_id
62 GROUP BY p.product_category_name
63 HAVING MAX(oi.price) - MIN(oi.price) > 500
64 ORDER BY price_difference DESC;
65
```

Data Output Messages Notifications

	product_category_name character varying (100)	price_difference numeric
1	utilidades_domesticas	6731.94
2	pcs	6694.50
3	artes	6495.50
4	eletroportateis	4792.50
5	instrumentos_musicais	4394.97
6	consoles_games	4094.81
7	esporte_lazer	4054.50
8	relogios_presentes	3990.91
9	[null]	3977.00
10	ferramentas_jardim	3923.65
11	bebes	3895.46
12	informatica_acessorios	3696.09
13	beleza_saude	3122.80
14	cool_stuff	3102.99
15	construcao_ferramentas_seguranca	3091.00
16	industria_comercio_e_negocios	3061.10
17	agro_industria_e_comercio	2977.01
18	portateis_casa_forno_e_cafe	2888.81

Products with Missing or Incomplete Category Names

BUSINESS CONTEXT

Clean, complete catalog data underpins search, filtering and recommendation quality across the platform.

KEY FINDINGS

- 614 products carry a missing or unusable category name.
- The large majority are NULL values rather than single-character entries.
- The issue is concentrated in specific product batches, not spread evenly.

BUSINESS INSIGHT

- Uncategorized SKUs are effectively invisible to category browsing and search, directly limiting discoverability and revenue.
- This is a data-governance gap rather than a demand problem — a low-cost, high-leverage fix.

RECOMMENDATION

- Run a one-time catalog remediation pass in India using auto-classification from product title and description.
- Enforce mandatory category tagging at seller onboarding to prevent recurrence.

Query Query History

```
77
78 -- Retrieve products where the product category name
79 -- is missing or contains only a single character.
80
81 SELECT
82     product_id,
83     product_category_name
84 FROM amazon_brazil.product
85 WHERE product_category_name IS NULL
86     OR LENGTH(TRIM(product_category_name)) = 1;
87
88
```

Data Output Messages Notifications

	product_id [PK] character varying (50)	product_category_name character varying (100)
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]
8	794de06c32a626a5692ff50e4985d3...	[null]
9	7af3e2da474486a3519b0cba9dea8...	[null]
10	629beb8e7317703dcc5f35b5463fd2...	[null]
11	3a78f64aac654298e4b9aff32fc21818	[null]
12	bcb815bba008d89458e428078c0b9...	[null]
13	6b82874c6b51b92913dccb364eaaa...	[null]
14	c68b419d9c6038271b85bac98adb0...	[null]
15	1dcd65bb5dd967d7b4c6b0223cefb...	[null]
16	671446e8e3aa3df1eca47b6c354a2...	[null]
17	f0ea71b6e2ab4cb3bd8f5ba522a25a...	[null]
18	fedccbd5e370e8ddb7aae6fb4cb703...	[null]
19	212cc0fa7359ab242a697a03a574f7...	[null]
20	...	[null]

Total rows: 614 Query complete 00:00:01.233

Payment Type Usage by Order Value Segment

BUSINESS CONTEXT

Knowing which payment method dominates each spend tier informs checkout design and financing offers by segment.

KEY FINDINGS

- Low-value orders paid by credit card total 60,548 — the largest single combination.
- Medium-value credit card orders (15,303) and low-value boleto orders (16,444) form the next tier.
- High-value orders (>1,000 BRL) remain rare across every payment type combined.

BUSINESS INSIGHT

- Credit card is the default rail even for everyday low-value purchases, not just large ones — reliability here is business-critical.
- The high-ticket segment is a small but valuable niche warranting dedicated fraud and support attention.

RECOMMENDATION

- Prioritize low-friction, one-tap payment for the high-volume low-value segment.
- Layer buyer-protection assurances onto the smaller high-value order segment.

```
Query History
89 -- Segment order values into Low, Medium, and High.
90 -- Count the number of each payment type within each segment.
91 SELECT
92     CASE
93         WHEN payment_value < 200 THEN 'Low (<200 BRL)'
94         WHEN payment_value BETWEEN 200 AND 1000 THEN 'Medium (200-1000 BRL)'
95         ELSE 'High (>1000 BRL)'
96     END AS order_value_segment,
97     payment_type,
98     COUNT(*) AS count
99 FROM amazon_brazil.payments
100 GROUP BY
101     CASE
102         WHEN payment_value < 200 THEN 'Low (<200 BRL)'
103         WHEN payment_value BETWEEN 200 AND 1000 THEN 'Medium (200-1000 BRL)'
104         ELSE 'High (>1000 BRL)'
105     END,
106     payment_type
107 ORDER BY count DESC;
108
```

Data Output Messages Notifications

	order_value_segment text	payment_type character varying (30)	count bigint
1	Low (<200 BRL)	credit_card	60548
2	Medium (200-1000 B...	voucher	286
3	High (>1000 BRL)	debit_card	15
4	Medium (200-1000 B...	boleto	3162
5	Medium (200-1000 B...	debit_card	227
6	Low (<200 BRL)	debit_card	1287
7	High (>1000 BRL)	boleto	178
8	High (>1000 BRL)	credit_card	944
9	Low (<200 BRL)	not_defined	3
10	Medium (200-1000 B...	credit_card	15303
11	Low (<200 BRL)	voucher	5476
12	High (>1000 BRL)	voucher	13
13	Low (<200 BRL)	boleto	16444

2 Price Range and Average Price by Category

BUSINESS CONTEXT

Category-level pricing benchmarks guide assortment and pricing strategy ahead of new-market entry.

KEY FINDINGS

- pcs leads at an average of 1,098.34 BRL, spanning 34.50–6,729.00 BRL.
- portateis_casa_forno_e_cafe (624.29) and eletrodomesticos_2 (476.12) follow.
- Lifestyle categories like musica (158.80) sit at the affordable end.

BUSINESS INSIGHT

- Electronics and appliance categories anchor the top of the price ladder, while lifestyle categories anchor the affordable end.
- The wide range within top categories confirms both entry-level and premium buyers shop the same category.

RECOMMENDATION

- Lead India's category launch with high-AOV electronics and appliances to build margin.
- Pair with affordable lifestyle categories to drive traffic and purchase frequency.

```
Query  Query History
109
110 -- Calculate the minimum, maximum, and average price
111 -- for each product category and sort by average price.
112
113 SELECT
114     p.product_category_name,
115     MIN(oi.price) AS min_price,
116     MAX(oi.price) AS max_price,
117     ROUND(AVG(oi.price), 2) AS avg_price
118 FROM amazon_brazil.product p
119 INNER JOIN amazon_brazil.order_items oi
120 ON p.product_id = oi.product_id
121 GROUP BY p.product_category_name
122 ORDER BY avg_price DESC;
123
```

Data Output Messages Notifications

	product_category_name character varying (100)	min_price numeric	max_price numeric	avg_price numeric
1	pcs	34.50	6729.00	1098.34
2	portateis_casa_forno_e_cafe	10.19	2899.00	624.29
3	eletrodomesticos_2	13.90	2350.00	476.12
4	agro_industria_e_comercio	12.99	2990.00	341.66
5	instrumentos_musicais	4.90	4399.87	281.62
6	eletroportateis	6.50	4799.00	280.78
7	portateis_cozinha_e_preparadores_de_aliment...	17.42	1099.00	264.57
8	telefonias_fixas	6.00	1790.00	225.69
9	construcao_ferramentas_seguranca	8.90	3099.90	208.99
10	relogios_presentes	8.99	3999.90	200.91
11	climatizacao	10.90	1599.00	185.27
12	moveis_quarto	6.90	650.00	183.75
13	pc_gamer	129.99	239.00	171.77
14	cool_stuff	7.00	3109.99	167.36
15	moveis_cozinha_area_de_servico_jantar_e_jar...	9.60	1320.00	164.87
16	moveis_escritorio	25.00	1189.90	162.01
17	musica	3.85	1165.97	158.80

Customers with More Than One Order

BUSINESS CONTEXT

Repeat purchasers underpin customer lifetime value, making them the foundation of any retention strategy.

KEY FINDINGS

- The most active repeat customers each placed 16 total orders, the observed ceiling.
- A distinct cohort of customers reaches this ceiling, rather than a single outlier.
- Order counts drop off quickly beyond the top repeat-buyer cohort.

BUSINESS INSIGHT

- A ceiling near 16 orders suggests even top buyers haven't been engaged through a formal loyalty relationship.
- This repeat-buyer cohort is a small slice of the overall customer base, reinforcing the retention gap seen in Q16.

RECOMMENDATION

- Design a tiered loyalty program that rewards customers as they cross order-count thresholds.
- Prioritize converting occasional buyers toward this repeat-buyer cohort as the primary growth lever.

Query Query History

```
124
125 -- Find customers who have placed more than one order.
126 -- Display their customer unique ID and total number of orders.
127
128 SELECT
129     c.customer_unique_id,
130     COUNT(o.order_id) AS total_orders
131 FROM amazon_brazil.customers c
132 INNER JOIN amazon_brazil.orders o
133 ON c.customer_id = o.customer_id
134 GROUP BY c.customer_unique_id
135 HAVING COUNT(o.order_id) > 1
136 ORDER BY total_orders DESC;
137
138
```

Data Output Messages Notifications

	customer_unique_id character varying (50)	total_orders bigint
1	a91e80fbe80ddc07de66a5cf927029...	16
2	a6168cd79131e64acef92e3c74d6cc...	16
3	363f980585bf04c1a88fdb986011c5...	16
4	cbd0350d4ccb9772e8e768d4a4a5...	16
5	417b909c0962b2610f1cfeb1c14789...	16
6	5f94af52aef02c968a2e0f01f430864e	16
7	1b6d29725255a77667a8c639eeb4c...	16
8	e4bbcc533fd3917c56dea2c43bf20...	16
9	930c4390af58f67334447c3a1cf2ba...	16
10	5bf4ea2d98005b960eea0dbf652ef4...	16
11	9159c04b88895d995741dd5b9b7a5...	16
12	4034aa08d48695a538b7030910aae...	16
13	c024307523462166b42112cfb6c8e...	16
14	0fdc0d21e1983e8af4d399e17671f7...	16
15	96fd69e8b0df76a9a807b01dc82bef...	16
16	7f4f709af2fd8fea44aacd30bca46264	16
17	f9c4e8531c2fe4159beb562fd7c2bd...	16

Customer Classification: New / Returning / Loyal

BUSINESS CONTEXT

A shared customer-type taxonomy (New / Returning / Loyal, by order count) enables consistent CRM segmentation.

KEY FINDINGS

- A CTE classifies every customer_unique_id by total order count.
- The output assigns a single customer_type label per customer across the full base.
- The large majority of sampled customers resolve to 'New', consistent with patterns seen elsewhere.

BUSINESS INSIGHT

- A CTE-based approach makes this segmentation reusable across marketing, CRM and reporting without rewriting logic.
- The dominance of 'New' customers confirms retention, not acquisition, is the larger growth lever.

RECOMMENDATION

- Replicate this CTE-based tagging in India's CRM and data warehouse from day one.
- Trigger automated second-purchase campaigns specifically targeted at the 'New' segment.

```
Query History
140
141 -- Categorize customers as New, Returning, or Loyal
142 -- based on the number of orders using a temporary table (CTE).
143
144 WITH customer_orders AS (
145     SELECT
146         c.customer_unique_id,
147         COUNT(o.order_id) AS order_count
148     FROM amazon_brazil.customers c
149     INNER JOIN amazon_brazil.orders o
150         ON c.customer_id = o.customer_id
151     GROUP BY c.customer_unique_id
152 )
153
154 SELECT
155     customer_unique_id,
156     CASE
157         WHEN order_count = 1 THEN 'New'
158         WHEN order_count BETWEEN 2 AND 4 THEN 'Returning'
159         ELSE 'Loyal'
160     END AS customer_type
161 FROM customer_orders
162 ORDER BY customer_unique_id;
163
164
```

Data Output Messages Notifications

	customer_unique_id character varying (50)	customer_type text
1	0000366f3b9a7992bf8c76cfd3221e2	New
2	0000b849f77a49e4a4ce2b2a4ca5be...	New
3	0000f46a3911fa3c08054444833370...	New
4	0000f6ccb0745a6a4b88665a16c9f0...	New
5	0004aac84e0df4da2b147fca70cf8255	New
6	0004bd2a26a76fe21f786e4fbd80607f	New
7	00050ab1314c0e55a6ca13cf7181fecf	New
8	00053a61a98854899e70ed204dd4b...	New
9	0005e1862207bf6ccc02e4228effd9a0	New

5 Top 5 Product Categories by Total Revenue

BUSINESS CONTEXT

Revenue concentration by category informs where Amazon India should focus sellers, ad inventory and warehousing.

KEY FINDINGS

- beleza_saude (health & beauty) leads at 1,257,865.34 BRL.
- relorios_presentes (1,203,060.32) and cama_mesa_banho (1,032,268.59) follow.
- esporte_lazer and informatica_acessorios round out the top five.

BUSINESS INSIGHT

- Health & beauty and gifting categories outperform pure electronics accessories, showing strong lifestyle and personal-care demand.
- The top 5 sit within a relatively narrow revenue band, indicating strength is spread across categories rather than one dominant winner.

RECOMMENDATION

- Prioritize health & beauty and home/gifting categories in India's initial seller-recruitment push.
- Balance this focus alongside electronics rather than treating either as the sole priority.

```
Query  Query History
166
167  -- Calculate the total revenue for each product category.
168  -- Display the top 5 product categories by total revenue.
169
170  SELECT
171      p.product_category_name,
172      ROUND(SUM(oi.price), 2) AS total_revenue
173  FROM  amazon_brazil.product p
174  INNER JOIN amazon_brazil.order_items oi
175      ON p.product_id = oi.product_id
176  GROUP BY p.product_category_name
177  ORDER BY total_revenue DESC
178  LIMIT 5;
179
180
181
```

Data Output Messages Notifications

	product_category_name character varying (100)	total_revenue numeric
1	beleza_saude	1257865.34
2	relorios_presentes	1203060.32
3	cama_mesa_banho	1032268.59
4	esporte_lazer	985881.10
5	informatica_acessorios	910605.07

Total Sales by Season

BUSINESS CONTEXT

Seasonal demand patterns guide campaign calendars and inventory pre-positioning across the year.

KEY FINDINGS

- Spring (4,934,990.20 BRL) and Summer (4,890,902.19 BRL) are essentially tied for the top season.
- Winter follows at 3,416,284.67 BRL.
- Autumn is the weakest season at 2,766,695.06 BRL.

BUSINESS INSIGHT

- Spring and Summer together drive the clear majority of annual revenue, reinforcing the mid-year peak identified in Q4.
- Autumn's wide gap below other seasons marks it as a natural window for promotional stimulus.

RECOMMENDATION

- Schedule India's flagship sale events to reinforce the natural mid-year demand peak.
- Test a dedicated stimulus campaign for the Autumn-equivalent low period before committing budget.

Query Query History

```
181
182 -- Calculate total sales for each season
183 -- using a subquery based on order purchase dates.
184
185 SELECT
186     season,
187     ROUND(SUM(payment_value), 2) AS total_sales
188 FROM
189     (
190         SELECT
191             p.payment_value,
192             CASE
193                 WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN (3,4,5) THEN 'Spring'
194                 WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN (6,7,8) THEN 'Summer'
195                 WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN (9,10,11) THEN 'Autumn'
196                 ELSE 'Winter'
197             END AS season
198         FROM amazon_brazil.orders o
199         INNER JOIN amazon_brazil.payments p
200             ON o.order_id = p.order_id
201     ) AS seasonal_sales
202 GROUP BY season
203 ORDER BY total_sales DESC;
204
205
206
```

Data Output Messages Notifications



	season text	total_sales numeric
1	Spring	4934990.20
2	Summer	4890902.19
3	Winter	3416284.67
4	Autumn	2766695.06

2 Products Selling Above Average Quantity

BUSINESS CONTEXT

Identifying volume outperformers supports reorder planning and best-seller merchandising decisions.

KEY FINDINGS

- The top product sold 527 units, the highest in the dataset.
- The next tier follows closely at 488, 484 and 392 units.
- A clear top tier sells meaningfully more than the above-average threshold itself.

BUSINESS INSIGHT

- A small set of hero SKUs carries disproportionate volume, a classic long-tail sales distribution.
- These hero products are strong candidates for guaranteed stock and negotiated seller terms.

RECOMMENDATION

- Build a best-seller guarantee program in India offering priority restocking for top-volume SKUs.
- Apply this same above-average filtering logic to continuously refresh the featured-placement list

```
207 -- Find products whose total quantity sold
208 -- is greater than the overall average quantity sold.
209
210 SELECT
211     product_id,
212     total_quantity_sold
213 FROM
214     (
215         SELECT
216             product_id,
217             COUNT(*) AS total_quantity_sold
218         FROM amazon_brazil.order_items
219         GROUP BY product_id
220     ) product_sales
221 WHERE total_quantity_sold >
222
223     SELECT AVG(product_count)
224     FROM
225         (
226             SELECT COUNT(*) AS product_count
227             FROM amazon_brazil.order_items
228             GROUP BY product_id
229         ) avg_sales
230
231 ORDER BY total_quantity_sold DESC;
232
233
```

Data Output Messages Notifications

	product_id character varying (50)	total_quantity_sold bigint
1	aca2eb7d00ea1a7b8ebd4e6831466...	527
2	99a4788cb24856965c36a24e339b6...	488
3	422879e10f46682990de24d770e7f...	484
4	389d119b48cf3043d311335e499d9...	392
5	368c6c730842d78016ad823897a37...	388
6	53759a2ecddad2bb87a079a1f1519f...	373
7	d1c427060a0f73f6b889a5c7c61f2a...	343

Monthly Revenue Trend — 2018

BUSINESS CONTEXT

The finance team needs visibility into peak and low revenue periods across 2018 to plan cash flow and campaigns.

KEY FINDINGS

- Revenue holds steady between 990K and 1.16M BRL per month from January through August.
- September (4,439.54) and October (589.67) collapse sharply below any prior month.

BUSINESS INSIGHT

- The Jan–Aug plateau confirms stable, predictable demand rather than a declining business.
- The Sep–Oct drop aligns with the dataset's known capture cutoff and should be flagged, not treated as a real sales collapse.

RECOMMENDATION

- Validate data completeness (max order date) before any trend reporting.
- Use the stable Jan–Aug pattern, not the Sep–Oct dip, as the baseline for India revenue planning.

Query Query History

```

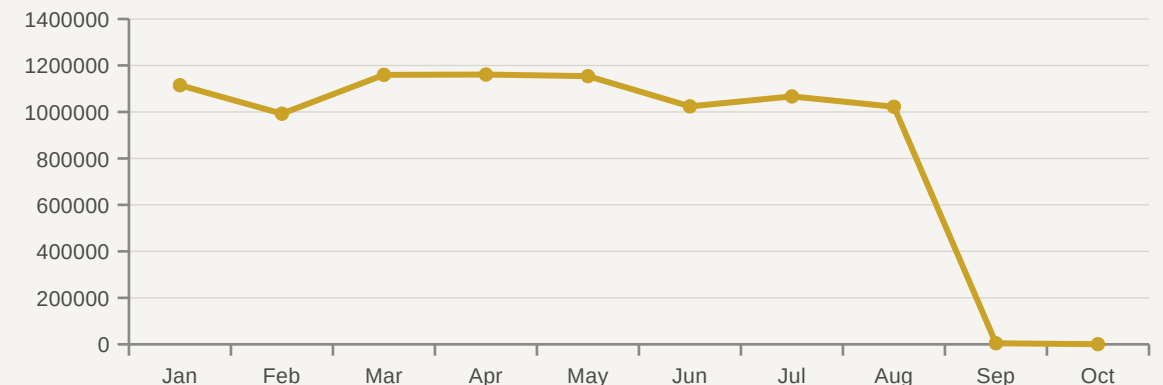
235 -- Calculate total monthly revenue for the year 2018.
236
237
238
239
240 SELECT
241     EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month,
242     ROUND(SUM(p.payment_value), 2) AS total_revenue
243 FROM amazon_brazil.orders o
244 INNER JOIN amazon_brazil.payments p
245     ON o.order_id = p.order_id
246 WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
247 GROUP BY EXTRACT(MONTH FROM o.order_purchase_timestamp)
248 ORDER BY month;
249
250

```

Data Output Messages Notifications

	month numeric	total_revenue numeric
1	1	1115004.18
2	2	992463.34
3	3	1159652.12
4	4	1160785.48
5	5	1153982.15
6	6	1023880.50
7	7	1066540.75
8	8	1022425.32
9	9	4439.54
10	10	589.67

2018 Monthly Revenue (BRL)



Loyalty Segmentation: Occasional / Regular / Loyal

BUSINESS CONTEXT

An effective loyalty program requires a clear purchase-frequency segmentation to target rewards efficiently.

KEY FINDINGS

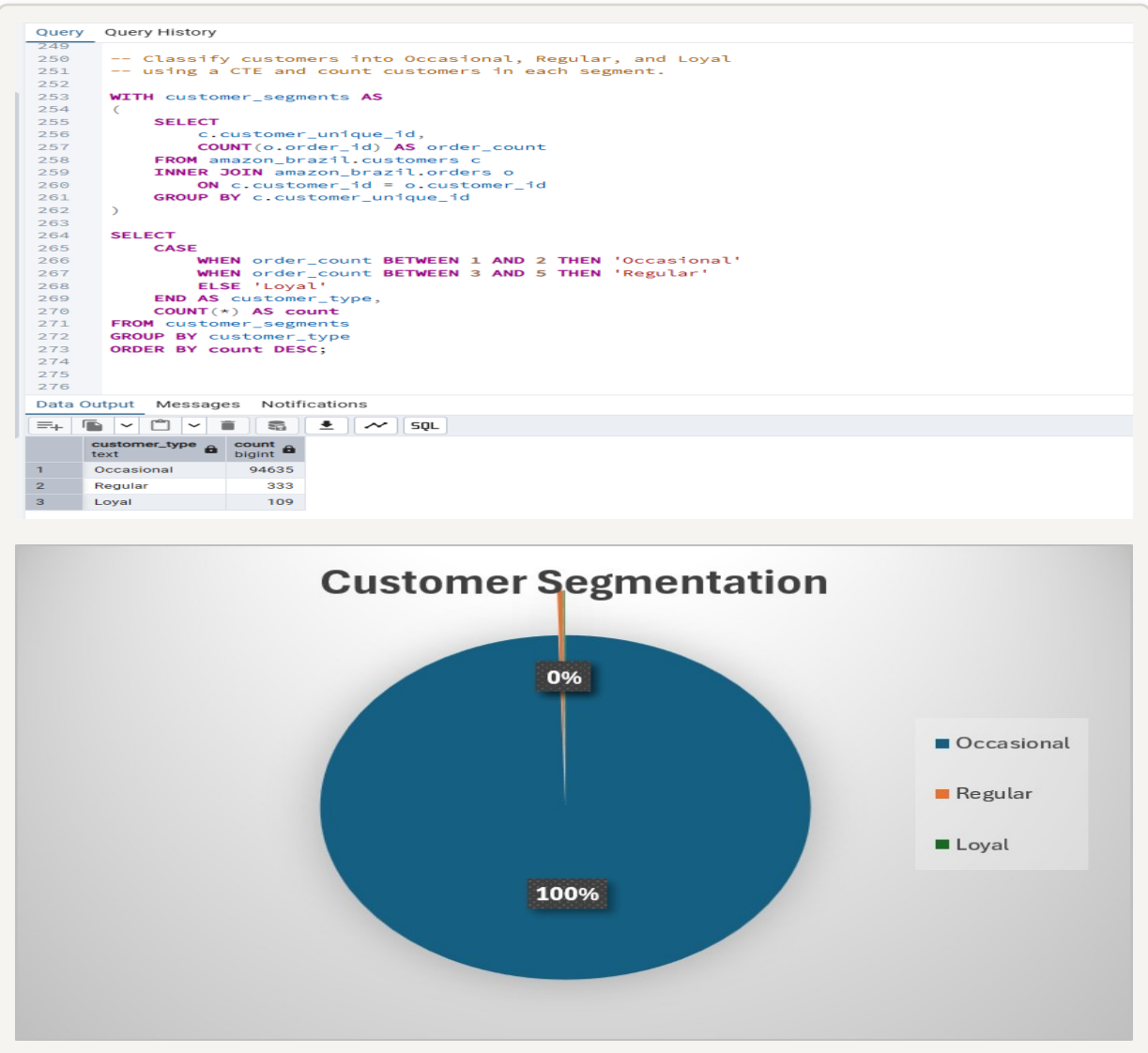
- Occasional customers (1-2 orders) total 94,635.
- Regular customers (3-5 orders) total just 333.
- Loyal customers (>5 orders) total only 109.

BUSINESS INSIGHT

- Occasional buyers make up approximately 99.5% of the customer base, showing repeat purchasing has not been cultivated at scale.
- The combined Regular and Loyal segments (under 450 customers) form a tiny, high-value pool worth protecting closely.

RECOMMENDATION

- Make first-to-second-purchase conversion the top CRM priority for Amazon India at launch.
- Track this Occasional-to-Regular transition rate as a core retention KPI from day one.



Top 20 Customers by Average Order Value

BUSINESS CONTEXT

Amazon India seeks to identify high-value customers to seed an exclusive rewards program.

KEY FINDINGS

- The top-ranked customer averages 13,664.08 BRL per order, over 5x the second-highest.
- Ranks 2–5 cluster between 6,726 and 7,275 BRL.
- Values taper more gradually from rank 6 onward, down to roughly 4,513 BRL by rank 11.

BUSINESS INSIGHT

- Using RANK() correctly preserves tied values, ensuring the VIP list isn't arbitrarily truncated.
- The top customer is a clear outlier, warranting individual account management rather than generic tier treatment.

RECOMMENDATION

- Launch an invite-only VIP tier in India seeded by this same ranking methodology.
- Assign dedicated servicing to outlier top-spend customers identified via RANK().

```

Query  Query History
275
276 -- Rank customers based on their average order value.
277 -- Display the top 20 customers.
278
279 WITH customer_avg AS
280 (
281     SELECT
282         o.customer_id,
283         ROUND(AVG(p.payment_value), 2) AS avg_order_value
284     FROM amazon_brazil.orders o
285     INNER JOIN amazon_brazil.payments p
286         ON o.order_id = p.order_id
287     GROUP BY o.customer_id
288 )
289
290 SELECT
291     customer_id,
292     avg_order_value,
293     RANK() OVER (ORDER BY avg_order_value DESC) AS customer_rank
294 FROM customer_avg
295 ORDER BY customer_rank
296 LIMIT 20;
297
  
```

Data Output Messages Notifications

	customer_id character varying (50)	avg_order_value numeric	customer_rank bigint
1	1617b1357756262bfa56ab541c47...	13664.08	1
2	ec5b2ba62e574342386871631fafd...	7274.88	2
3	c6e2731c5b391845f6800c97401a4...	6929.31	3
4	f48d464a0baaea338cb25f816991a...	6922.21	4
5	3fd6777bbce08a352fdd04e4a7cc...	6726.66	5
6	05455dfa7cd02f13d132aa7a6a972...	6081.54	6
7	df55c14d1476a9a3467f131269c24...	4950.34	7
8	e0a2412720e9ea4f26c1ac985f6a7...	4809.44	8
9	24bbf5fd2f2e1b359ee7de94defc4a...	4764.34	9
10	3d979689f636322c62418b6346b1...	4681.78	10
11	1afc82cd60e303ef09b4ef9837c95...	4513.32	11

Monthly Cumulative Sales by Product

BUSINESS CONTEXT

Tracking cumulative sales per product from its first sale reveals product lifecycle and growth trajectory.

KEY FINDINGS

- A window function accumulates monthly sales into a running total per product_id, ordered chronologically.
- Products show varied trajectories — some single-month spikes, others gradual monthly build-up.
- The running total resets logically at each product's first recorded sale month.

BUSINESS INSIGHT

- A running-total view distinguishes sustained, compounding demand from one-off sales spikes — separating durable sellers from fads.
- This lifecycle lens is more actionable than a single lifetime total, since it captures momentum, not just size.

RECOMMENDATION

- Adopt cumulative sales tracking as a standard product-health metric in India.
- Use momentum trends to trigger reorder points and flag products that have plateaued.

```

300
301 -- Calculate monthly cumulative sales for each product.
302
303 WITH monthly_sales AS
304 (
305     SELECT
306         oi.product_id,
307         DATE_TRUNC('month', o.order_purchase_timestamp)::date AS sale_month,
308         SUM(oi.price) AS monthly_sales
309     FROM amazon_brazil.order_items oi
310     INNER JOIN amazon_brazil.orders o
311         ON oi.order_id = o.order_id
312     GROUP BY
313         oi.product_id,
314         DATE_TRUNC('month', o.order_purchase_timestamp)::date
315 )
316
317 SELECT
318     product_id,
319     sale_month,
320     SUM(monthly_sales) OVER
321     (
322         PARTITION BY product_id
323         ORDER BY sale_month
324     ) AS total_sales
325 FROM monthly_sales
326 ORDER BY product_id, sale_month;

```

Data Output Messages Notifications

	product_id character varying (50)	sale_month date	total_sales numeric
1	00066f42aeeb9f3007548bb9d3f33c38	2018-05-01	101.65
2	00088930e925c41fd95ebfe695fd2655	2017-12-01	129.90
3	0009406fd7479715e4bef61dd91f2462	2017-12-01	229.00
4	000b8f95fcb9e0096488278317764d...	2018-08-01	117.80
5	000d9be29b5207b54e86aa1b1ac54...	2018-04-01	199.00
6	0011c512eb256aa0dbbb544d8dfcf6e	2017-12-01	52.00
7	00126f27c813603687e6ce486d909d...	2017-09-01	498.00

Monthly Sales & MoM Growth by Payment Type (2018)

BUSINESS CONTEXT

Amazon India wants to see how each payment method's sales evolve month over month to inform payments strategy.

KEY FINDINGS

- Boleto sales move from 204,844.66 BRL in January to 183,112.72 in February (−10.61%).
- March through May stay near-flat, with growth of +4.60%, +1.05%, and +0.95%.
- June drops sharply to 153,350.28 BRL, a −21.51% swing.

BUSINESS INSIGHT

- The LAG() window function turns month-over-month growth into a reusable, first-class metric rather than a manual calculation.
- Boleto's real volatility (−21.5% to +4.6%) contrasts with flatter card-based methods, suggesting boleto users are more cash-flow sensitive.

RECOMMENDATION

- Track MoM growth by payment method as a standing dashboard metric in India.
- Test targeted incentives on the more volatile payment rails to help smooth demand.

Query 1: Calculate monthly sales and month-over-month growth for each payment type in the year 2018.

```
330 -- Calculate monthly sales and month-over-month growth
331 -- for each payment type in the year 2018.
332
333 WITH monthly_sales AS
334 (
335     SELECT
336         p.payment_type,
337         DATE_TRUNC('month', o.order_purchase_timestamp)::date AS sale_month,
338         SUM(p.payment_value) AS monthly_total
339     FROM amazon_brazil.orders o
340     INNER JOIN amazon_brazil.payments p
341         ON o.order_id = p.order_id
342     WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
343     GROUP BY
344         p.payment_type,
345         DATE_TRUNC('month', o.order_purchase_timestamp)::date
346 )
347
348 SELECT
349     payment_type,
350     sale_month,
351     ROUND(monthly_total, 2) AS monthly_total,
352     ROUND(
353         (
354             monthly_total -
355             LAG(monthly_total) OVER
356             (
357                 PARTITION BY payment_type
358                 ORDER BY sale_month
359             )
360         ) /
361         LAG(monthly_total) OVER
362             (
363                 PARTITION BY payment_type
364                 ORDER BY sale_month
365             )
366         ) * 100,
367         2) AS monthly_change
368 FROM monthly_sales
369 ORDER BY payment_type, sale_month;
```

Data Output:

payment_type	sale_month	monthly_total	monthly_change
boleto	2018-01-01	204844.66	[null]
boleto	2018-02-01	183112.72	-10.61
boleto	2018-03-01	191538.02	4.60
boleto	2018-04-01	193547.09	1.05
boleto	2018-05-01	195378.93	0.95
boleto	2018-06-01	153350.28	-21.51

Query 2: Calculate monthly sales and month-over-month growth for each payment type in the year 2018.

```
355 monthly_total -
356 LAG(monthly_total) OVER
357 (
358     PARTITION BY payment_type
359     ORDER BY sale_month
360 )
361 /
362 NULLIF(
363     LAG(monthly_total) OVER
364     (
365         PARTITION BY payment_type
366         ORDER BY sale_month
367     ),
368     0
369 ) * 100,
370 2) AS monthly_change
371 FROM monthly_sales
372 ORDER BY payment_type, sale_month;
```

Data Output:

payment_type	sale_month	monthly_total	monthly_change
boleto	2018-01-01	204844.66	[null]
boleto	2018-02-01	183112.72	-10.61
boleto	2018-03-01	191538.02	4.60
boleto	2018-04-01	193547.09	1.05
boleto	2018-05-01	195378.93	0.95
boleto	2018-06-01	153350.28	-21.51

Overall Findings

Customer Behavior

- Credit card drives 73.9% of orders — the default rail even for everyday, low-value purchases.
- A small repeat-buyer cohort tops out near 16 orders, well below a formal loyalty ceiling.

Payment Analysis

- Card-based methods carry the widest transaction variability; voucher is the most consistent.
- Every order-value segment is credit-card dominant, concentrating processing risk.

Product Performance

- Health & beauty and gifting categories lead total revenue generation.
- A small set of hero SKUs (500+ units) carries disproportionate sales volume.

Revenue Trends

- Spring and Summer together drive the clear majority of annual revenue.
- Demand peaks span May–August — a sustained season, not one holiday event.

Customer Retention

- 99.5% of customers are 'Occasional' (1–2 orders) — the single biggest growth gap.
- Regular and Loyal segments combined are under 450 customers — a high-value niche.

Window Function Insights

- RANK() and LAG() cleanly surface VIP customers and month-over-month growth trends.
- Cumulative-sales tracking distinguishes durable sellers from one-off spikes.

Business Recommendations

1

Customer Retention

Target the 'New' segment with automated second-purchase campaigns, evidence-based on the 99.5% Occasional finding (Q11, Q16).

2

Repeat Purchases

Pilot a tiered loyalty program rewarding order-count milestones, seeded from the 16-order top-repeat-buyer cohort (Q10).

3

Pricing Strategy

Evaluate Good-Better-Best tiers in categories with the widest price spread, such as home goods and PCs (Q5, Q9).

4

Category Prioritization

Prioritize seller recruitment in health & beauty, gifting and home categories, which led total revenue (Q12).

5

Payment Optimization

Validate India's actual payment mix before designing checkout, rather than assuming Brazil's card-first pattern applies (Q2, Q8).

6

Marketing Timing

Align flagship sale events with a locally-verified demand peak, informed by Brazil's May–August pattern (Q4, Q13).

7

Inventory Planning

Use cumulative sales tracking and above-average volume filters to guide reorder points for momentum products (Q14, Q18).

8

Operational Efficiency

Remediate the 614 products with missing category data and standardize CTE-based segmentation across CRM tools (Q7, Q11).

Thank You

Amazon Brazil Sales Data Analysis — SQL Milestone Project

Sahil Makkar | NextLeap Data Analytics Fellowship